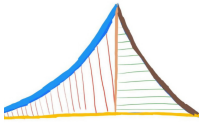
Math for EE



G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. CBSE CLASS 10 MATHEMATICS (2025, SET 2)

A. Formulae

1.1.1 Tangent–radius property. If PA is a tangent to a circle with centre O at the point A,

$$OA \perp PA \tag{1.1.1.1}$$

1.1.2 Tangents from an external point. If PA and PB are tangents from P to a circle with centre O,

$$PA = PB, \quad \triangle OAP \cong \triangle OBP, \quad \angle POA = \angle POB \tag{1.1.2.1}$$

Solution: $OA = OB$ (radii), $PA = PB$ and OP is common, so the triangles are congruent by SSS.

1.1.3 Angle sum of a quadrilateral and of a triangle,

$$\sum \text{angles} = 360^\circ \text{ (quadrilateral)}, \quad 180^\circ \text{ (triangle)} \tag{1.1.3.1}$$

1.1.4 Angle in a semicircle. If AB is a diameter and C lies on the circle,

$$\angle ACB = 90^\circ \tag{1.1.4.1}$$

1.1.5 Tangent–chord theorem. The angle between a tangent and a chord through the point of contact equals the angle in the alternate segment. If PC is tangent at C,

$$\angle PCB = \angle CAB \tag{1.1.5.1}$$

1.1.6 AA similarity. If two angles of one triangle equal two angles of another, the triangles are similar and corresponding sides are in the same ratio. For $\triangle LMN \sim \triangle ABC$,

$$\frac{LM}{AB} = \frac{MN}{BC} = \frac{LN}{AC} \tag{1.1.6.1}$$

1.1.7 Section formula. The point $\mathbf{R}$ dividing $\mathbf{P}$ and $\mathbf{Q}$ internally in the ratio $m : n$,

$$\mathbf{R} = \frac{m\mathbf{Q} + n\mathbf{P}}{m+n} = \mathbf{P} + \frac{m}{m+n}(\mathbf{Q} - \mathbf{P}) \tag{1.1.7.1}$$

For $m = n$ this is the mid-point $\frac{\mathbf{P}+\mathbf{Q}}{2}$.

1.1.8 Squared length and perpendicularity,

$$\|\mathbf{A}\|^2 = \mathbf{A}^\top \mathbf{A}, \quad \mathbf{A} \perp \mathbf{B} \iff \mathbf{A}^\top \mathbf{B} = 0 \tag{1.1.8.1}$$

1.1.9 Angle between two vectors. For direction vectors $\mathbf{m}_1$ and $\mathbf{m}_2$,

$$\cos\theta = \frac{\mathbf{m}_1^\top \mathbf{m}_2}{\|\mathbf{m}_1\| \, \|\mathbf{m}_2\|} \tag{1.1.9.1}$$

1.1.10 Altitude to the hypotenuse. If $\angle BAC = 90^\circ$ and $AD \perp BC$, then

$$AD^2 = BD \cdot DC \tag{1.1.10.1}$$

Solution: Take $\mathbf{B} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\mathbf{D} = \begin{pmatrix} BD \\ 0 \end{pmatrix}$, $\mathbf{C} = \begin{pmatrix} BD + DC \\ 0 \end{pmatrix}$, $\mathbf{A} = \begin{pmatrix} BD \\ h \end{pmatrix}$. From

(1.1.8.1),

$$(\mathbf{B}-\mathbf{A})^{\top}(\mathbf{C}-\mathbf{A}) = -BD\cdot DC + h^2 = 0 \tag{1.1.10.2}$$

1.1.11 Zeroes of a quadratic. For ax^2+bx+c with zeroes α,β,

$$\alpha+\beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a} \tag{1.1.11.1}$$

1.1.12 Newton–Raphson method. To solve $f(x)=0$,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \tag{1.1.12.1}$$

Solution: The tangent to f at x_n is $y = f(x_n)+f'(x_n)(x-x_n)$. Setting $y=0$ gives x_{n+1}.

1.1.13 Arithmetic progression with first term a and common difference d,

$$a_n = a+(n-1)d, \quad S_n = \frac{n}{2}[2a+(n-1)d] = \frac{n}{2}(a+a_n) \tag{1.1.13.1}$$

1.1.14 Gaussian elimination and rank. For a system $\mathbf{Ax}=\mathbf{b}$ with n unknowns, row-reduce the augmented matrix $\begin{pmatrix}\mathbf{A} & \mathbf{b}\end{pmatrix}$. The system is consistent if and only if

$$\operatorname{rank}(\mathbf{A}) = \operatorname{rank}\begin{pmatrix}\mathbf{A} & \mathbf{b}\end{pmatrix} \tag{1.1.14.1}$$

and the solution is unique when this common rank equals n.

1.1.15 Simple interest on a principal P at rate $r\%$ per annum for t years,

$$I = \frac{Prt}{100} \tag{1.1.15.1}$$

1.1.16 Standard values and inverse sine,

$$\sin 60^\circ = \frac{\sqrt{3}}{2}, \quad \tan 30^\circ = \frac{1}{\sqrt{3}}, \quad \tan 45^\circ = 1 \tag{1.1.16.1}$$

The principal value $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = 60^\circ$.

1.1.17 Angle of depression. From $\mathbf{B}$ at height h above the origin, the line of sight to a ground point with angle of depression θ has direction $\mathbf{m} = \begin{pmatrix}\cos\theta \\ -\sin\theta\end{pmatrix}$ and meets the ground $\mathbf{e}_2^{\top}\mathbf{x}=0$ at

$$\mathbf{B}+\lambda\mathbf{m}, \quad \lambda = -\frac{\mathbf{e}_2^{\top}\mathbf{B}}{\mathbf{e}_2^{\top}\mathbf{m}} = \frac{h}{\sin\theta} \tag{1.1.17.1}$$

1.1.18 Unit step, convolution and the z-transform. The unit step is $u[n]=1$ for $n\geq 0$ and 0 otherwise, and

$$(x*h)[n] = \sum_k x[k]\,h[n-k], \quad X(z) = \sum_n x[n]\,z^{-n} \tag{1.1.18.1}$$

$$(x*h)[n] \overset{\mathcal{Z}}{\longleftrightarrow} X(z)\,H(z) \tag{1.1.18.2}$$

The running sum of a sequence is its convolution with $u[n]$.

1.1.19 Probability by convolution. For a pmf $p[n]$ and an event with indicator $m[n] \in \{0,1\}$, let $h[n] = m[-n]$. Then

$$\Pr(E) = \sum_n p[n]\; m[n] = (p * h)\;[0] \tag{1.1.19.1}$$

which is the coefficient of z^0 in $P(z)\,H(z)$. Also $\Pr(E) + \Pr(E') = 1$.

1.1.20 Grouped data. For a class with lower limit l, width h, frequency f, preceding cumulative frequency cf, and class mark x_i,

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} \tag{1.1.20.1}$$

$$\text{Median} = l + \frac{\frac{N}{2} - cf}{f} \times h, \quad N = \sum f_i \tag{1.1.20.2}$$

The modal class is the class with the highest frequency.

B. Problems

1.2.1 If tangents PA and PB drawn from an external point P to a circle with centre O are inclined to each other at an angle of 80°, then $\angle POA$ is equal to (CBSE 2025 Set 2, Q.1)

a) 40° b) 50° c) 60° d) 80°

Solution: Take the centre $\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ as the origin and $\mathbf{P} = \begin{pmatrix} d \\ 0 \end{pmatrix}$ on the x-axis. Let r be the radius and $\phi = \angle POA$. The points of contact are symmetric about OP, so

$$\mathbf{A} = r\begin{pmatrix} \cos\phi \\ \sin\phi \end{pmatrix}, \quad \mathbf{B} = r\begin{pmatrix} \cos\phi \\ -\sin\phi \end{pmatrix} \tag{1.2.1.1}$$

The radius is perpendicular to the tangent at the point of contact,

$$(\mathbf{A}-\mathbf{O})^\top(\mathbf{A}-\mathbf{P}) = 0 \implies \|\mathbf{A}\|^2 = \mathbf{A}^\top\mathbf{P} \implies r^2 = rd\cos\phi \implies r = d\cos\phi \tag{1.2.1.2}$$

Hence the direction vectors of the tangents PA and PB are

$$\mathbf{A}-\mathbf{P} = d\begin{pmatrix} \cos^2\phi - 1 \\ \sin\phi\cos\phi \end{pmatrix} = d\sin\phi\begin{pmatrix} -\sin\phi \\ \cos\phi \end{pmatrix} \tag{1.2.1.3}$$

$$\mathbf{B}-\mathbf{P} = d\begin{pmatrix} \cos^2\phi - 1 \\ -\sin\phi\cos\phi \end{pmatrix} = d\sin\phi\begin{pmatrix} -\sin\phi \\ -\cos\phi \end{pmatrix} \tag{1.2.1.4}$$

The unit direction vectors are $\mathbf{m}_1 = \begin{pmatrix} -\sin\phi \\ \cos\phi \end{pmatrix}$ and $\mathbf{m}_2 = \begin{pmatrix} -\sin\phi \\ -\cos\phi \end{pmatrix}$, so the angle between the tangents satisfies

$$\cos\angle APB = \mathbf{m}_1^\top\mathbf{m}_2 = \sin^2\phi - \cos^2\phi = -\cos 2\phi \tag{1.2.1.5}$$

Given $\angle APB = 80°$,

$$\cos 2\phi = -\cos 80° = \cos 100° \implies \phi = 50° \implies \angle POA = 50° \quad \text{(Option B)} \tag{1.2.1.6}$$

Taking $r = 2$, we get $d = \frac{r}{\cos 50°} \approx 3.11$ and the points

$$\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{P} = \begin{pmatrix} 3.11 \\ 0 \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 1.29 \\ 1.53 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1.29 \\ -1.53 \end{pmatrix} \tag{1.2.1.7}$$

The following code verifies the solution

codes/1.c

The following code generates Fig. 1.2.1.1.

codes/1plot.py

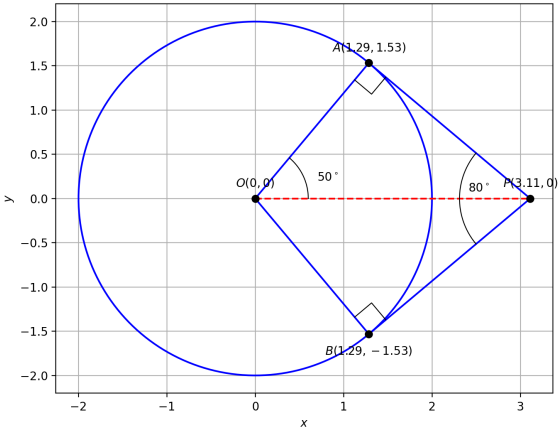


Fig. 1.2.1.1: Tangents PA and PB plotted from the coordinates of $\mathbf{O}$, $\mathbf{P}$, $\mathbf{A}$, $\mathbf{B}$

1.2.2 Two vertices of $\triangle PQR$ are $P(-1, 5)$ and $Q(5, 2)$. The coordinates of the point R on PQ which divides PQ in the ratio $2 : 1$ are (CBSE 2025 Set 2, Q.4)

a) $(3, -3)$ b) $(5, 5)$ c) $(3, 3)$ d) $(5, 1)$

Solution: Let $\mathbf{P} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$ and $\mathbf{Q} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$. The displacement from P to Q is

$$\mathbf{Q} - \mathbf{P} = \begin{pmatrix} 6 \\ -3 \end{pmatrix} \tag{1.2.2.1}$$

Since $PR : RQ = 2 : 1$, the point R lies two-thirds of the way from P to Q:

$$\mathbf{R} = \mathbf{P} + \frac{2}{3}(\mathbf{Q} - \mathbf{P}) = \begin{pmatrix} -1 \\ 5 \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} \quad \text{(Option C)} \tag{1.2.2.2}$$

The following code verifies the solution

codes/4.c

The following code generates Fig. 1.2.2.1.

codes/4plot.py

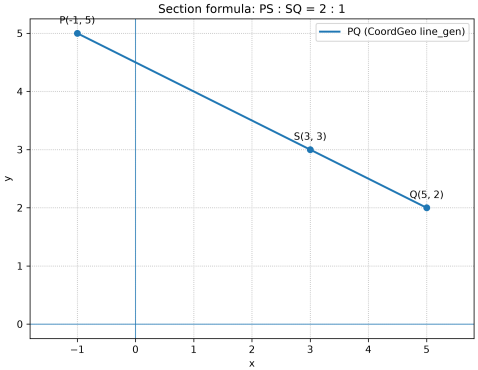


Fig. 1.2.2.1: Point R dividing PQ in the ratio $2 : 1$

1.2.3 If $\sin 4\theta = \frac{\sqrt{3}}{2}$, then $\frac{\theta}{3}$ is equal to (CBSE 2025 Set 2, Q.5)

a) $60°$ b) $20°$ c) $15°$ d) $5°$

Solution: Taking the principal (acute) value,

$$4\theta = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) = 60° \implies \theta = 15° \implies \frac{\theta}{3} = 5° \quad \text{(Option D)} \tag{1.2.3.1}$$

The following code verifies the solution

codes/5.c

1.2.4 The zeroes of the polynomial $p(y) = 7y^2 - \frac{11}{3}y - \frac{2}{3}$ are (CBSE 2025 Set 2, Q.6)

a) $-\frac{2}{3}, -\frac{1}{7}$ b) $-\frac{2}{7}, -\frac{1}{3}$ c) $\frac{2}{3}, \frac{1}{7}$ d) $\frac{2}{3}, -\frac{1}{7}$

Solution: Multiplying by 3, the zeroes of p are the zeroes of

$$g(y) = 21y^2 - 11y - 2, \quad g'(y) = 42y - 11 \tag{1.2.4.1}$$

The Newton–Raphson iteration is

$$y_{n+1} = y_n - \frac{g(y_n)}{g'(y_n)} = y_n - \frac{21y_n^2 - 11y_n - 2}{42y_n - 11} \tag{1.2.4.2}$$

Starting from $y_0 = 0.7$ and $y_0 = -0.2$, the iterations converge to

$$y = \frac{2}{3}, \quad y = -\frac{1}{7} \quad \text{(Option D)} \tag{1.2.4.3}$$

which agree with the factorisation $g(y) = (3y-2)(7y+1)$. The following code verifies the solution

codes/6.c

The following code generates Fig. 1.2.4.1.

codes/6plot.py

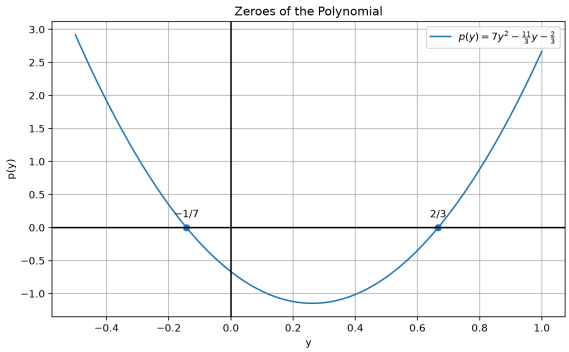


Fig. 1.2.4.1: Graph of $p(y)$ and its zeroes

1.2.5 $AOBC$ is a rectangle whose three vertices are $A(0,2)$, $O(0,0)$ and $B(4,0)$. The square of the length of its diagonal is (CBSE 2025 Set 2, Q.8)

a) 36 b) 20 c) 16 d) 4

Solution: The fourth vertex is $C(4,2)$ and the diagonal is AB. With $\mathbf{A} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}$,

$$\mathbf{B} - \mathbf{A} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}, \qquad \|\mathbf{B} - \mathbf{A}\|^2 = (\mathbf{B} - \mathbf{A})^\top (\mathbf{B} - \mathbf{A}) = 16 + 4 = 20 \quad \text{(Option B)} \tag{1.2.5.1}$$

The following code verifies the solution

codes/8.c

The following code generates Fig. 1.2.5.1.

codes/8plot.py

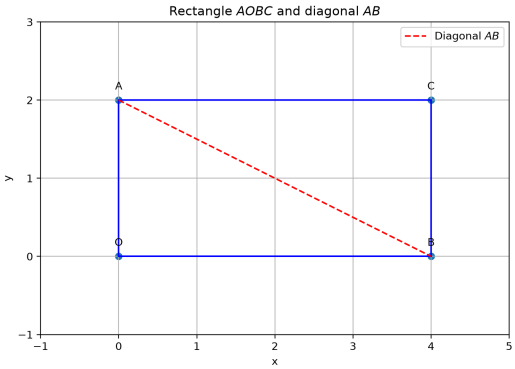


Fig. 1.2.5.1: Rectangle $AOBC$ and its diagonals

1.2.6 A die is thrown once. The probability of getting a number which is not a factor of 36 is (CBSE 2025 Set 2, Q.10)

a) $\frac{1}{2}$ b) $\frac{2}{3}$ c) $\frac{1}{6}$ d) $\frac{5}{6}$

Solution: Let the outcome of the throw be n. The probability mass function of a fair die is

$$p[n] = \frac{1}{6}\left(u[n-1] - u[n-7]\right) \tag{1.2.6.1}$$

$$\implies P(z) = \sum_{n} p[n] z^{-n} = \frac{1}{6}\left(z^{-1} + z^{-2} + \cdots + z^{-6}\right) \tag{1.2.6.2}$$

where $u[n]$ is the unit step. Define the indicator of the event E ("not a factor of 36"). Among $1, \ldots, 6$ only 5 is not a factor of 36, so

$$m[n] = \delta[n-5], \quad M(z) = z^{-5} \tag{1.2.6.3}$$

and $\Pr(E) = \sum_n p[n]\, m[n]$. With the reversed sequence $h[n] = m[-n] = \delta[n+5]$, whose transform is $H(z) = z^5$, the convolution at $n = 0$ gives exactly this sum:

$$(p * h)[0] = \sum_{k} p[k]\; h[-k] = \sum_{k} p[k]\; m[k] = \Pr(E) \tag{1.2.6.4}$$

Convolution becomes multiplication in the z-domain,

$$Y(z) = P(z)\,H(z) = \frac{1}{6}\sum_{n=1}^{6} z^{5-n} = \frac{1}{6}\left(z^4 + z^3 + z^2 + z + 1 + z^{-1}\right) \quad (1.2.6.5)$$

and $(p * h)\,[0]$ is the coefficient of z^0. Therefore,

$$\Pr(E) = \frac{1}{6} \quad \text{(Option C)} \quad (1.2.6.6)$$

The following code verifies the solution

codes/10.c

1.2.7 In the figure, $\angle BAC = 90^\circ$ and $AD \perp BC$. If $BC = 16$ cm and $DC = 4$ cm, then $AC = x$ is equal to (CBSE 2025 Set 2, Q.11)

a) 4 cm b) 5 cm c) 8 cm d) 3 cm

Solution: Take $\mathbf{B} = \begin{pmatrix}0\\0\end{pmatrix}$, $\mathbf{C} = \begin{pmatrix}16\\0\end{pmatrix}$. Then $BD = 16 - 4 = 12$, so $\mathbf{D} = \begin{pmatrix}12\\0\end{pmatrix}$ and, since $AD \perp BC$, $\mathbf{A} = \begin{pmatrix}12\\h\end{pmatrix}$. As $\angle BAC = 90^\circ$,

$$(\mathbf{B} - \mathbf{A})^\top (\mathbf{C} - \mathbf{A}) = 0 \quad (1.2.7.1)$$

$$\begin{pmatrix}-12 & -h\end{pmatrix}\begin{pmatrix}4\\-h\end{pmatrix} = -48 + h^2 = 0 \implies h^2 = 48 \quad (1.2.7.2)$$

Hence,

$$x^2 = \|\mathbf{C} - \mathbf{A}\|^2 = 4^2 + h^2 = 64 \implies x = 8 \text{ cm} \quad \text{(Option C)} \quad (1.2.7.3)$$

The following code verifies the solution

codes/11.c

The following code generates Fig. 1.2.7.1.

codes/11plot.py

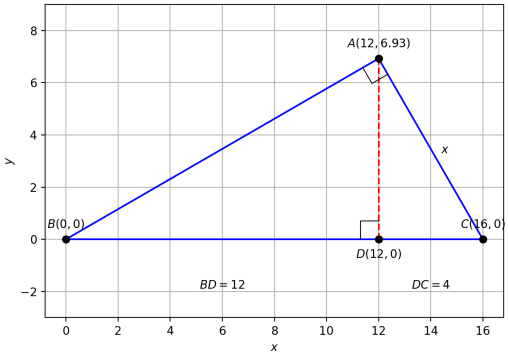


Fig. 1.2.7.1: Right triangle ABC with altitude AD, plotted from the coordinates of **B**, **C**, **D**, **A**

1.2.8 The mid-point of the line segment joining $(a, 4)$ and $(2, 2b)$ is $(2, 6)$. Then $a + b$ is equal to (CBSE 2025 Set 2, Q.15)

a) 6 b) 7 c) 8 d) 16

Solution: The mid-point is the average of the position vectors:

$$\begin{pmatrix} 2 \\ 6 \end{pmatrix} = \frac{1}{2}\left(\begin{pmatrix} a \\ 4 \end{pmatrix} + \begin{pmatrix} 2 \\ 2b \end{pmatrix}\right) \implies \frac{a+2}{2} = 2, \quad \frac{4+2b}{2} = 6 \implies a = 2, \quad b = 4 \tag{1.2.8.1}$$

Therefore $a + b = 6$ (Option A). The following code verifies the solution

codes/15.c

The following code generates Fig. 1.2.8.1.

codes/15plot.py

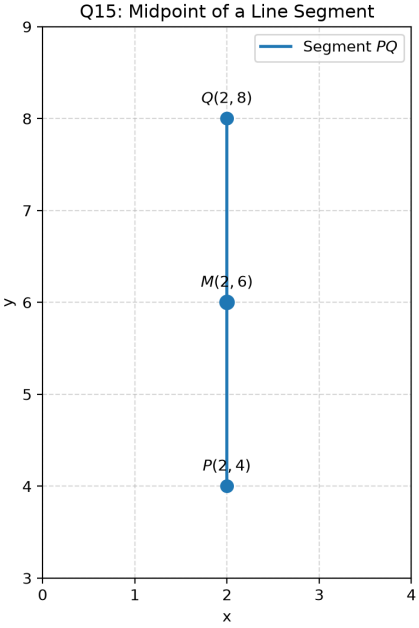


Fig. 1.2.8.1: Segment PQ and its mid-point M

1.2.9 $\angle M = \angle B = 130^\circ$ and $\angle N = 28^\circ$. If $LM = 45$ cm, $MN = 63$ cm, $LN = 72$ cm and $AB = 5$ cm, then AC is equal to (CBSE 2025 Set 2, Q.17)

a) 16 cm b) 7 cm c) 8 cm d) 4 cm

Solution: In $\triangle LMN$, $\angle L = 180^\circ - 130^\circ - 28^\circ = 22^\circ = \angle A$. Since $\angle A = \angle L$ and $\angle B = \angle M$, by AA similarity

$$\triangle LMN \sim \triangle ABC \quad (L \leftrightarrow A,\ M \leftrightarrow B,\ N \leftrightarrow C) \tag{1.2.9.1}$$

The scale factor is $k = \frac{AB}{LM} = \frac{5}{45} = \frac{1}{9}$, so

$$AC = k \cdot LN = \frac{72}{9} = 8 \text{ cm} \quad \text{(Option C)} \tag{1.2.9.2}$$

The following code verifies the solution

codes/17.c

The following code generates Fig. 1.2.9.1. The triangle LMN is drawn from the given angles with $LN = 72$ cm, and ABC is its image under the scale factor $\frac{1}{9}$; the figure is not to scale with respect to MN.

codes/17plot.py

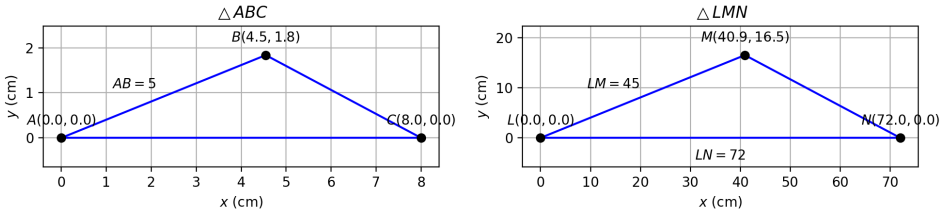


Fig. 1.2.9.1: Triangles ABC and LMN plotted from coordinates

1.2.10 $\triangle DEF$ and $\triangle PQR$ are such that $\angle D = \angle Q$ and $\angle E = \angle R$. Which of the following proportions is <u>not</u> true? (CBSE 2025 Set 2, Q.18)

a) $\frac{DE}{QR} = \frac{DF}{PQ}$ b) $\frac{EF}{RP} = \frac{DF}{PQ}$ c) $\frac{EF}{RP} = \frac{DE}{QR}$ d) $\frac{DE}{PQ} = \frac{EF}{RP}$

Solution: By AA similarity, $\triangle DEF \sim \triangle QRP$ with $D \leftrightarrow Q$, $E \leftrightarrow R$, $F \leftrightarrow P$. Corresponding sides are proportional:

$$\frac{DE}{QR} = \frac{EF}{RP} = \frac{DF}{QP} \tag{1.2.10.1}$$

Options A, B and C compare corresponding sides. Option D compares DE with PQ, which are not corresponding sides. Hence the proportion that is not true is Option D. The following code checks all four options on a pair of similar triangles

codes/18.c

1.2.11 If α and β are the zeroes of the polynomial $x^2 - 2x - 1$, find the value of $\frac{1}{2\alpha} + \frac{1}{2\beta} + 3\alpha\beta$. (CBSE 2025 Set 2, Q.21(b))

Solution: Let $f(x) = x^2 - 2x - 1$, so $f'(x) = 2x - 2$. The Newton–Raphson iteration is

$$x_{n+1} = x_n - \frac{x_n^2 - 2x_n - 1}{2x_n - 2} \tag{1.2.11.1}$$

Starting from $x_0 = 2.5$ and $x_0 = -0.5$ gives

$$\alpha = 1 + \sqrt{2}, \quad \beta = 1 - \sqrt{2} \tag{1.2.11.2}$$

From the sum and product of zeroes, $\alpha + \beta = 2$ and $\alpha\beta = -1$, so

$$\frac{1}{2\alpha} + \frac{1}{2\beta} + 3\alpha\beta = \frac{\alpha + \beta}{2\alpha\beta} + 3\alpha\beta = \frac{2}{-2} - 3 = -4 \tag{1.2.11.3}$$

The following code verifies the solution

codes/21.c

The following code generates Fig. 1.2.11.1.

codes/21plot.py

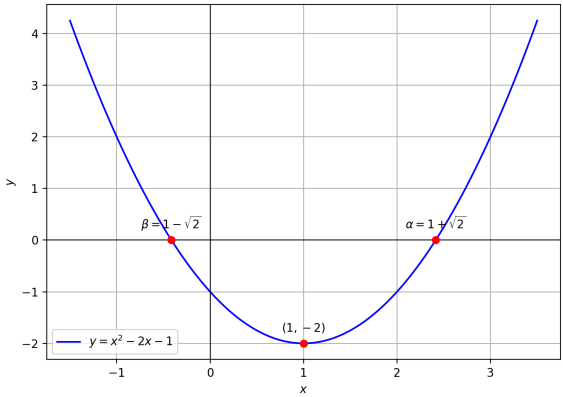


Fig. 1.2.11.1: Graph of $y = x^2 - 2x - 1$ with zeroes α and β

1.2.12 In a school, 65% of the students have black eyes, 25% have brown eyes and the rest have blue eyes. A student is chosen at random. Find the probability that the student has (a) blue eyes, (b) brown or black eyes. (CBSE 2025 Set 2, Q.27)

Solution: Let $n = 0, 1, 2$ denote black, brown and blue eyes. The probability of blue eyes is $1 - 0.65 - 0.25 = 0.10$, so

$$p[n] = 0.65\,\delta[n] + 0.25\,\delta[n-1] + 0.10\,\delta[n-2] \tag{1.2.12.1}$$

$$\implies P(z) = 0.65 + 0.25z^{-1} + 0.10z^{-2} \tag{1.2.12.2}$$

For an event with indicator $m[n]$, let $h[n] = m[-n]$. As in the previous problem,

$$\Pr(E) = \sum_k p[k]\; m[k] = (p * h)\;[0] \tag{1.2.12.3}$$

which is the coefficient of z^0 in $P(z)\,H(z)$.

(a) Blue eyes: $m[n] = \delta[n-2]$, so $h[n] = \delta[n+2]$ and $H(z) = z^2$,

$$P(z)\,H(z) = 0.65z^2 + 0.25z + 0.10 \tag{1.2.12.4}$$

$$\implies \Pr(\text{blue}) = 0.10 \tag{1.2.12.5}$$

(b) Brown or black eyes: $m[n] = \delta[n] + \delta[n-1]$, so $h[n] = \delta[n] + \delta[n+1]$ and

$$H(z) = 1 + z,$$

$$P(z)\,H(z) = 0.65z + 0.90 + 0.35z^{-1} + 0.10z^{-2} \tag{1.2.12.6}$$

$$\implies \Pr(\text{brown or black}) = 0.90 \tag{1.2.12.7}$$

The following code verifies the solution

codes/27.c

1.2.13 In the figure, PC is a tangent to the circle at C and AB is a diameter which, when extended, meets the tangent at P. If $\angle PCA = 110°$, find $\angle CBA$ and $\angle BCO$. (CBSE 2025 Set 2, Q.29)

Solution: Since AB is a diameter, $\angle ACB = 90°$. As P lies beyond B,

$$\angle PCA = \angle PCB + \angle BCA \implies \angle PCB = 110° - 90° = 20° \tag{1.2.13.1}$$

By the tangent–chord theorem, $\angle CAB = \angle PCB = 20°$. In $\triangle ABC$,

$$\angle CBA = 90° - 20° = 70° \tag{1.2.13.2}$$

Since $OB = OC$, $\triangle OBC$ is isosceles and

$$\angle BCO = \angle CBO = 70° \tag{1.2.13.3}$$

The following code verifies the solution

codes/29.c

The following code generates Fig. 1.2.13.1.

codes/29plot.py

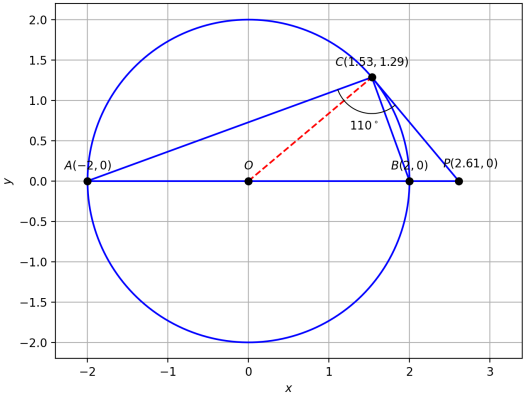


Fig. 1.2.13.1: Tangent PC and diameter AB, plotted from the coordinates of **A**, **B**, **C**, **P**

1.2.14 Find the coordinates of the points which divide the line segment joining $A\,(-2,2)$ and $B\,(2,8)$ into four equal parts. (CBSE 2025 Set 2, Q.31(b))

Solution: With $\mathbf{A} = \begin{pmatrix} -2 \\ 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 \\ 8 \end{pmatrix}$, $\mathbf{B} - \mathbf{A} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$. The division points are $\mathbf{A} + \frac{k}{4}(\mathbf{B} - \mathbf{A})$ for $k = 1, 2, 3$:

$$\mathbf{P} = \begin{pmatrix} -1 \\ 3.5 \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} 1 \\ 6.5 \end{pmatrix} \tag{1.2.14.1}$$

The following code verifies the solution

codes/31.c

The following code generates Fig. 1.2.14.1.

codes/31plot.py

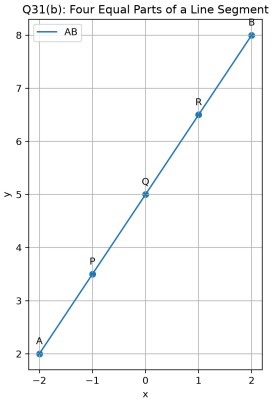


Fig. 1.2.14.1: Division of AB into four equal parts

1.2.15 The nth term of an AP is 4, its common difference is 2 and the sum of its first n terms is -14. Find n, the first term, and the sum of the first 20 terms. (CBSE 2025 Set 2, Q.32(a))

Solution: Let $u\,[n]$ be the unit step function, $u\,[n] = 1$ for $n \geq 0$ and 0 otherwise. Index the terms from $n = 0$, so that the AP with first term a and common difference d is the causal sequence

$$x\,[n] = (a + nd)\; u\,[n] \tag{1.2.15.1}$$

with $x\,[0] = a$ and $x\,[n] = 0$ for $n < 0$. Since $u\,[n-k] = 1$ only for $k \leq n$, the

running sum is the convolution of x with the unit step,

$$y[n] = (x * u)[n] = \sum_k x[k]\, u[n-k] = x[0] + x[1] + \cdots + x[n] \tag{1.2.15.2}$$

$$= (n+1)\,a + \frac{n(n+1)}{2} d \tag{1.2.15.3}$$

for $n \geq 0$. Let m be the index of the last term, so that the number of terms is $m+1$. Here $d = 2$, $x[m] = 4$ and $y[m] = -14$. From $x[m] = a + 2m = 4$ we get $a = 4 - 2m$, and

$$y[m] = (m+1)(4-2m) + m(m+1) = (m+1)(4-m) = -14 \tag{1.2.15.4}$$

$$\implies m^2 - 3m - 18 = 0 \implies (m-6)(m+3) = 0 \tag{1.2.15.5}$$

Since $m \geq 0$, $m = 6$, so the number of terms is $m+1 = 7$ and $a = 4 - 12 = -8$. The sum of the first 20 terms is $y[19]$,

$$y[19] = 20(-8) + \frac{19 \cdot 20}{2}(2) = -160 + 380 = 220 \tag{1.2.15.6}$$

The following code verifies the solution

codes/32a.c

1.2.16 A man lent part of his money at 10% per annum and the rest at 15% per annum. His yearly interest was ₹1900. Had he interchanged the two rates, he would have earned ₹200 more. Find the two amounts. (CBSE 2025 Set 2, Q.34)

Solution: Let x and y be the amounts lent at 10% and 15%. Then

$$0.10x + 0.15y = 1900 \tag{1.2.16.1}$$

$$0.15x + 0.10y = 2100 \tag{1.2.16.2}$$

The augmented matrix of $\mathbf{Ax} = \mathbf{b}$ is $\begin{pmatrix}\mathbf{A} & \mathbf{b}\end{pmatrix}$. Using Gaussian elimination (Gauss–Jordan row reduction),

$$\begin{pmatrix} 0.10 & 0.15 & 1900 \\ 0.15 & 0.10 & 2100 \end{pmatrix} \xleftrightarrow{R_1 \leftarrow 10R_1} \begin{pmatrix} 1 & 1.5 & 19000 \\ 0.15 & 0.10 & 2100 \end{pmatrix} \tag{1.2.16.3}$$

$$\xleftrightarrow{R_2 \leftarrow R_2 - 0.15R_1} \begin{pmatrix} 1 & 1.5 & 19000 \\ 0 & -0.125 & -750 \end{pmatrix} \tag{1.2.16.4}$$

$$\xleftrightarrow{R_2 \leftarrow R_2/(-0.125)} \begin{pmatrix} 1 & 1.5 & 19000 \\ 0 & 1 & 6000 \end{pmatrix} \tag{1.2.16.5}$$

$$\xleftrightarrow{R_1 \leftarrow R_1 - 1.5R_2} \begin{pmatrix} 1 & 0 & 10000 \\ 0 & 1 & 6000 \end{pmatrix} \tag{1.2.16.6}$$

Both $\mathbf{A}$ and $\begin{pmatrix}\mathbf{A} & \mathbf{b}\end{pmatrix}$ have two non-zero rows, so

$$\text{rank}(\mathbf{A}) = \text{rank}\begin{pmatrix}\mathbf{A} & \mathbf{b}\end{pmatrix} = 2 = \text{number of unknowns} \tag{1.2.16.7}$$

Hence the system is consistent with a unique solution,

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 10000 \\ 6000 \end{pmatrix} \tag{1.2.16.8}$$

The amounts are ₹10000 at 10% and ₹6000 at 15%. The following code verifies the solution

codes/34.c

1.2.17 **Case study (rainfall).** The distribution of the number of meteorological subdivisions by annual rainfall is given below. (CBSE 2025 Set 2, Q.36)

Rainfall (mm)	200–400	400–600	600–800	800–1000	1000–1200	1200–1400
Subdivisions	3	4	7	4	3	3

TABLE 1.2.17: Rainfall distribution, total 24 subdivisions

(i) Find the modal class. (ii) Find the median rainfall **or** the mean rainfall. (iii) Find the number of subdivisions with rainfall of at least 800 mm.

Solution: **(i)** The highest frequency is 7, so the modal class is 600–800 mm.

(ii) Here $N = 24$ and $\frac{N}{2} = 12$. The cumulative frequencies are $3, 7, 14, 18, 21, 24$, so the median class is 600–800 with $l = 600$, $cf = 7$, $f = 7$, $h = 200$:

$$\text{Median} = l + \frac{\frac{N}{2} - cf}{f} \times h = 600 + \frac{12 - 7}{7} \times 200 \approx 742.86 \text{ mm} \tag{1.2.17.1}$$

For the mean, with class marks $x_i = 300, 500, \ldots, 1300$,

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{18600}{24} = 775 \text{ mm} \tag{1.2.17.2}$$

(iii) The classes from 800 mm upward have $4 + 3 + 3 = 10$ subdivisions.

The following code verifies the solution

codes/36.c

1.2.18 **Case study (lighthouse).** A lighthouse AB is 50 m high. From its top B, the angles of depression of a ship at P and, after some time, at Q are 30° and 45° respectively, with P, Q, A on the same line. (CBSE 2025 Set 2, Q.38)

Solution: Take the foot of the lighthouse as the origin, $\mathbf{A} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 \\ 50 \end{pmatrix}$, with the ground along the x-axis, $\mathbf{e}_2^\top \mathbf{x} = 0$. The line of sight from $\mathbf{B}$ with angle of depression θ has direction vector $\mathbf{m} = \begin{pmatrix} \cos\theta \\ -\sin\theta \end{pmatrix}$ and meets the ground when

$$\mathbf{e}_2^\top (\mathbf{B} + \lambda \mathbf{m}) = 0 \implies \lambda = -\frac{\mathbf{e}_2^\top \mathbf{B}}{\mathbf{e}_2^\top \mathbf{m}} = \frac{50}{\sin\theta} \tag{1.2.18.1}$$

For $\mathbf{Q}$ ($\theta = 45°$) and $\mathbf{P}$ ($\theta = 30°$),

$$\mathbf{Q} = \mathbf{B} + 50\sqrt{2} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 50 \\ 0 \end{pmatrix} \tag{1.2.18.2}$$

$$\mathbf{P} = \mathbf{B} + 100 \cdot \frac{1}{2} \begin{pmatrix} \sqrt{3} \\ -1 \end{pmatrix} = \begin{pmatrix} 50\sqrt{3} \\ 0 \end{pmatrix} \tag{1.2.18.3}$$

(i) The distance QA is

$$QA = \|\mathbf{Q} - \mathbf{A}\| = 50 \text{ m} \tag{1.2.18.4}$$

(iii)(a) The distance moved from P to Q is

$$PQ = \|\mathbf{P} - \mathbf{Q}\| = 50\left(\sqrt{3} - 1\right) \approx 36.60 \text{ m} \tag{1.2.18.5}$$

(iii)(b) The ship covers $QA = 50$ m $= 0.05$ km in 10 minutes $= \frac{1}{6}$ h, so

$$\text{Speed} = \frac{0.05}{1/6} = 0.3 \text{ km/h} \tag{1.2.18.6}$$

The following code verifies the solution

codes/38.c

The following code generates Fig. 1.2.18.1.

codes/38plot.py

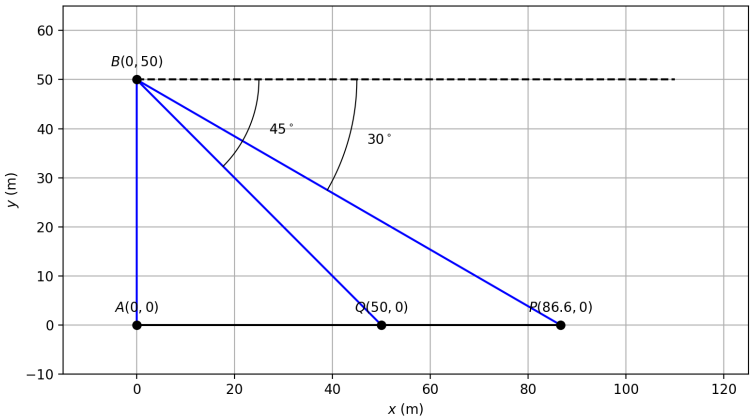


Fig. 1.2.18.1: Lighthouse AB and ship positions P, Q, plotted from coordinates

2. CBSE CLASS 12 MATHEMATICS (65/7/1)

A. Formulae

2.1.1 Domain of inverse functions. $\cos^{-1} x$ and $\sin^{-1} x$ are defined only for $-1 \le x \le 1$, while $\sin x$ and $\cos x$ are defined for all real x. The domain of a sum is the intersection of the domains.

2.1.2 Counting. If each of m independent positions can take c values, the total number of arrangements is

$$N = c^m \tag{2.1.2.1}$$

2.1.3 Symmetric matrix. A square matrix is symmetric if

$$\mathbf{A}^\top = \mathbf{A} \tag{2.1.3.1}$$

Every diagonal matrix is symmetric.

2.1.4 Determinant properties. For $n \times n$ matrices $\mathbf{A}$, $\mathbf{B}$ and a scalar k,

$$|k\mathbf{A}| = k^n |\mathbf{A}|, \qquad |\mathbf{AB}| = |\mathbf{A}|\,|\mathbf{B}| \tag{2.1.4.1}$$

2.1.5 Adjugate. With $\mathbf{A}\,\mathrm{adj}\,\mathbf{A} = |\mathbf{A}|\,\mathbf{I}$,

$$|\mathrm{adj}\,\mathbf{A}| = |\mathbf{A}|^{n-1} \tag{2.1.5.1}$$

Solution: Taking determinants of $\mathbf{A}\,\mathrm{adj}\,\mathbf{A} = |\mathbf{A}|\,\mathbf{I}$ gives $|\mathbf{A}|\,|\mathrm{adj}\,\mathbf{A}| = |\mathbf{A}|^n$.

2.1.6 Characteristic polynomial and Cayley–Hamilton theorem. The characteristic polynomial of a square matrix is $f(\lambda) = |\mathbf{A} - \lambda\mathbf{I}|$, and

$$|\mathbf{A}| = f(0) = \lambda_1 \cdots \lambda_n, \qquad f(\mathbf{A}) = \mathbf{0} \tag{2.1.6.1}$$

For a 2×2 matrix, $\mathbf{A}^2 - \mathrm{tr}\,(\mathbf{A})\,\mathbf{A} + |\mathbf{A}|\,\mathbf{I} = \mathbf{0}$; the eigenvalues sum to the trace and multiply to the determinant. For a 3×3 matrix with $f(\lambda) = -\lambda^3 + s_1\lambda^2 - s_2\lambda + s_3$, Cayley–Hamilton gives

$$\mathrm{adj}\,\mathbf{A} = \mathbf{A}^2 - s_1\mathbf{A} + s_2\mathbf{I} \tag{2.1.6.2}$$

2.1.7 Cramer's rule. For $\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} e \\ f \end{pmatrix}$ with $\Delta = ad - bc \neq 0$,

$$x = \frac{ed - bf}{\Delta}, \quad y = \frac{af - ec}{\Delta} \tag{2.1.7.1}$$

2.1.8 Monotonicity. If $f'(x) > 0$ for all x in an interval, f is strictly increasing there; if $f'(x) < 0$, it is strictly decreasing.

2.1.9 Logarithm and exponential. For $x > 0$,

$$e^{k \log x} = x^k \tag{2.1.9.1}$$

2.1.10 Integral properties.

$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx, \qquad \int_{-a}^a f(x)\,dx = \begin{cases} 2\int_0^a f(x)\,dx & f \text{ even} \\ 0 & f \text{ odd} \end{cases} \tag{2.1.10.1}$$

$$\int_0^{2a} f(x)\,dx = \int_0^a f(x)\,dx + \int_0^a f(a+x)\,dx \tag{2.1.10.2}$$

2.1.11 Trapezoidal rule. With n subintervals of width $h = \frac{b-a}{n}$ and nodes $x_i = a + ih$,

$$\int_a^b f(x)\,dx \approx T_n = \frac{h}{2}\left(f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\right) \tag{2.1.11.1}$$

The error is $-\frac{(b-a)h^2}{12} f''(\xi)$ for some $\xi \in (a,b)$, so T_n converges to the integral as $n \to \infty$.

2.1.12 Standard integrals,

$$\int \frac{du}{\sqrt{1-u^2}} = \sin^{-1} u + C, \qquad \int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C \tag{2.1.12.1}$$

2.1.13 Derivatives. Product rule, chain rule and the exponential rule,

$$\frac{d}{dx}(uv) = u'v + uv', \quad \frac{d}{dx}a^x = a^x \ln a, \quad \frac{d}{dx}\log(\sin u) = \cot u \frac{du}{dx} \tag{2.1.13.1}$$

2.1.14 Symmetric integral. If $g(\pi - x) = g(x)$,

$$\int_0^\pi x\,g(x)\,dx = \frac{\pi}{2}\int_0^\pi g(x)\,dx \tag{2.1.14.1}$$

Solution: Substituting $x \to \pi - x$ gives $I = \int_0^\pi (\pi - x)\,g(x)\,dx$. Adding this to the original, $2I = \pi \int_0^\pi g(x)\,dx$.

2.1.15 Laplace transform. For a first-order equation $y' + ay = f(x)$,

$$\mathcal{L}\{y'\} = sY(s) - y(0), \quad \mathcal{L}\{\cos x\} = \frac{s}{s^2+1}, \quad \mathcal{L}\left\{e^{-ax}\right\} = \frac{1}{s+a} \tag{2.1.15.1}$$

$$\mathcal{L}\{1\} = \frac{1}{s}, \quad \mathcal{L}\{t\} = \frac{1}{s^2} \tag{2.1.15.2}$$

The order of a differential equation is the highest derivative present; the degree is its power.

2.1.16 Orthogonality and scalar product with a unit vector,

$$|\mathbf{a}+\mathbf{b}| = |\mathbf{a}-\mathbf{b}| \iff \mathbf{a}^\top\mathbf{b} = 0, \qquad \text{scalar product of } \mathbf{a} \text{ with unit vector along } \mathbf{v} = \frac{\mathbf{a}^\top}{\|\mathbf{v}} \tag{2.1.16.1}$$

2.1.17 Distance between parallel lines $\mathbf{x} = \mathbf{P}_1 + \lambda\mathbf{d}$ and $\mathbf{x} = \mathbf{P}_2 + \mu\mathbf{d}$,

$$d = \frac{\|(\mathbf{P}_2 - \mathbf{P}_1) \times \mathbf{d}\|}{\|\mathbf{d}\|} \tag{2.1.17.1}$$

2.1.18 Points on a line in space. A line $\mathbf{x} = \mathbf{A} + t\mathbf{d}$ is written with one parameter t, and the distance of a point on it from $\mathbf{P}$ is $\|\mathbf{A} + t\mathbf{d} - \mathbf{P}\|$.

2.1.19 Foot of the perpendicular and image. For a point $\mathbf{P}$ and the line $\mathbf{A} + t\mathbf{d}$,

$$t_0 = \frac{(\mathbf{P} - \mathbf{A})^\top \mathbf{d}}{\mathbf{d}^\top \mathbf{d}}, \quad \mathbf{H} = \mathbf{A} + t_0\mathbf{d}, \quad \mathbf{P}' = 2\mathbf{H} - \mathbf{P} \tag{2.1.19.1}$$

2.1.20 Corner point theorem. The maximum or minimum of a linear objective over a feasible region occurs at a corner point. For an unbounded region, a minimum exists only if the objective is bounded below there.

2.1.21 Conditional probability,

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}, \qquad \Pr(A' \cap B) = \Pr(B) - \Pr(A \cap B) \tag{2.1.21.1}$$

and $\Pr(A \cap B) \leq \min\{\Pr(A), \Pr(B)\}$.

2.1.22 Probability by convolution and the z-transform. For a pmf $p[n]$ with $P(z) = \sum_n p[n] z^{-n}$, the pmf of the sum of independent variables is the convolution, $p_X * p_Y \leftrightarrow P_X(z) P_Y(z)$. For an event with indicator $m[n] \in \{0, 1\}$ and $h[n] = m[-n]$,

$$\Pr(E) = \sum_n m[n] p[n] = (p * h)[0], \qquad E[X] = \sum_n n p[n] = -\left.\frac{dP(z)}{dz}\right|_{z=1} \tag{2.1.22.1}$$

where $(p * h)[0]$ is the coefficient of z^0 in $P(z) H(z)$.

2.1.23 Total probability and Bayes' theorem. For mutually exclusive, exhaustive events $A_1, \ldots, A_n$,

$$\Pr(D) = \sum_i \Pr(D \mid A_i) \Pr(A_i), \qquad \Pr(A_1 \mid D) = \frac{\Pr(D \mid A_1) \Pr(A_1)}{\Pr(D)} \tag{2.1.23.1}$$

B. Problems

2.2.1 Find the domain of the function $f(x) = \cos^{-1} x + \sin x$. (CBSE Class 12, 65/7/1, Q.2)

Solution: $\cos^{-1} x$ is defined only for $-1 \leq x \leq 1$, and $\sin x$ is defined for every real x. The domain of the sum is the intersection of the two domains:

$$\text{Domain}(f) = [-1, 1] \quad \text{(Option C)} \tag{2.2.1.1}$$

The program scans x over $[-3, 3]$ and treats the points where `acos(x)+sin(x)` is not a number as outside the domain. The following code verifies the solution

codes/2.c

2.2.2 The number of all possible matrices of order 3×3 with each entry 2 or 3 is (CBSE Class 12, 65/7/1, Q.3)

Solution: Each of the 9 entries can independently be 2 or 3. By the multiplication principle,

$$N = 2^9 = 512 \quad \text{(Option B)} \tag{2.2.2.1}$$

The following code verifies the solution

codes/3.c

2.2.3 The matrix $\mathbf{A} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix}$ is a (CBSE Class 12, 65/7/1, Q.4)

a) scalar matrix b) identity matrix c) null matrix d) symmetric matrix

Solution: (theory only) A matrix is symmetric if $\mathbf{A}^\top = \mathbf{A}$. For a diagonal matrix the transpose leaves every entry unchanged,

$$\mathbf{A}^\top = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{pmatrix} = \mathbf{A} \tag{2.2.3.1}$$

The diagonal entries are not all equal, so $\mathbf{A}$ is not a scalar matrix; it is neither the identity nor the null matrix. Hence $\mathbf{A}$ is a symmetric matrix (Option D).

2.2.4 If $\mathbf{A}$ and $\mathbf{B}$ are square matrices of order 3 with $|\mathbf{A}| = 3$ and $|\mathbf{B}| = 5$, then $|2\mathbf{AB}|$ is (CBSE Class 12, 65/7/1, Q.5)

Solution: Following matgeo, the characteristic polynomial of a square matrix $\mathbf{M}$ is $f(\lambda) = |\mathbf{M} - \lambda\mathbf{I}|$, so

$$|\mathbf{M}| = f(0) = \lambda_1\lambda_2\lambda_3 \tag{2.2.4.1}$$

where λ_i are the eigenvalues. The eigenvalues of $2\mathbf{AB}$ are not those of $\mathbf{A}$ and $\mathbf{B}$ separately, so we use $|k\mathbf{A}| = k^n |\mathbf{A}|$ and $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$. With $n = 3$,

$$|2\mathbf{AB}| = 2^3 |\mathbf{A}| |\mathbf{B}| = 8(3)(5) = 120 \quad \text{(Option B)} \tag{2.2.4.2}$$

To check this numerically, take $\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}$ with $|\mathbf{A}| = 3$ and $\mathbf{B} = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix}$ with $|\mathbf{B}| = 5$. The program obtains the characteristic polynomial of $\mathbf{M} = 2\mathbf{AB}$ from `numpy.poly`, which returns $\det(\lambda\mathbf{I} - \mathbf{M})$, and reads $|\mathbf{M}| = (-1)^3 c_0$ from its constant term c_0. The following code verifies the solution

codes/5.py

2.2.5 If $\mathbf{A}$ is a square matrix of order 3 with $|\mathbf{A}| = 5$, then $|\text{adj}\,\mathbf{A}|$ is (CBSE Class 12, 65/7/1, Q.6)

Solution: With matgeo's characteristic polynomial $f(\lambda) = |\mathbf{A} - \lambda\mathbf{I}|$ of a 3×3 matrix,

$$f(\lambda) = -\lambda^3 + s_1\lambda^2 - s_2\lambda + s_3, \quad s_1 = \text{tr}(\mathbf{A}), \quad s_3 = |\mathbf{A}| \tag{2.2.5.1}$$

where s_2 is the sum of the principal 2×2 minors. By the Cayley–Hamilton theorem $f(\mathbf{A}) = \mathbf{0}$, so

$$\mathbf{A}\left(\mathbf{A}^2 - s_1\mathbf{A} + s_2\mathbf{I}\right) = s_3\mathbf{I} = |\mathbf{A}|\,\mathbf{I} \implies \text{adj}\,\mathbf{A} = \mathbf{A}^2 - s_1\mathbf{A} + s_2\mathbf{I} \tag{2.2.5.2}$$

If λ_i are the eigenvalues of $\mathbf{A}$, then $\text{adj}\,\mathbf{A} = |\mathbf{A}|\,\mathbf{A}^{-1}$ has eigenvalues $\mu_i = \frac{|\mathbf{A}|}{\lambda_i}$, and

$$|\text{adj}\,\mathbf{A}| = \mu_1\mu_2\mu_3 = \frac{|\mathbf{A}|^3}{\lambda_1\lambda_2\lambda_3} = |\mathbf{A}|^2 = 5^2 = 25 \quad \text{(Option C)} \tag{2.2.5.3}$$

The program takes $\mathbf{A} = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix}$ with $|\mathbf{A}| = 5$ and $f(\lambda)$ from `numpy.poly`, builds $\text{adj}\,\mathbf{A}$ using `polyvalm` from matgeo, and checks that $\mathbf{A}\,\text{adj}\,\mathbf{A} = 5\mathbf{I}$ and $|\text{adj}\,\mathbf{A}| = 25$. The following code verifies the solution

codes/6.py

2.2.6 The function $f(x) = 2x - \sin x$, $x \in \mathbb{R}$, is (CBSE Class 12, 65/7/1, Q.11)

Solution: Differentiating,

$$f'(x) = 2 - \cos x \tag{2.2.6.1}$$

Since $-1 \le \cos x \le 1$,

$$1 \le f'(x) \le 3 \implies f'(x) > 0 \quad \forall x \in \mathbb{R} \tag{2.2.6.2}$$

A differentiable function with a strictly positive derivative is strictly increasing, so f is strictly increasing on $\mathbb{R}$ (Option B). The program classifies each scanned point with `if-else` on the sign of $f'(x)$. The following code verifies the solution

codes/11.c

The following code generates Fig. 2.2.6.1.

codes/11plot.py

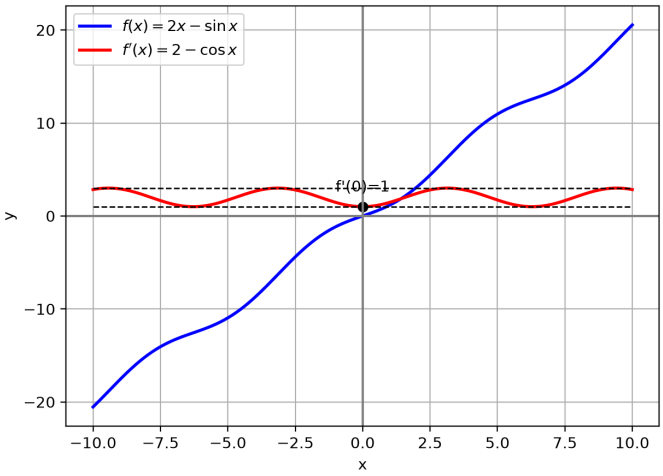


Fig. 2.2.6.1: $f(x)$ and $f'(x)$ with $1 \leq f'(x) \leq 3$

2.2.7 Evaluate $\int \frac{e^{9\log x} - e^{8\log x}}{e^{6\log x} - e^{5\log x}}\,dx$. (CBSE Class 12, 65/7/1, Q.12)

Solution: For $x > 0$, $e^{k\log x} = x^k$. Hence, for $x \neq 1$,

$$\frac{x^9 - x^8}{x^6 - x^5} = \frac{x^8(x-1)}{x^5(x-1)} = x^3 \tag{2.2.7.1}$$

$$\int x^3\,dx = \frac{x^4}{4} + C \quad \text{(Option C)} \tag{2.2.7.2}$$

To check this, apply the trapezoidal rule to the original integrand over $[2, 4]$. With $n = 4$ and $h = 0.5$ the nodes are $2, 2.5, 3, 3.5, 4$, where $f(x) = x^3$ takes the values $8, 15.625, 27, 42.875, 64$, so

$$T_4 = \frac{0.5}{2}\left(8 + 2\left(15.625 + 27 + 42.875\right) + 64\right) = 60.75 \tag{2.2.7.3}$$

which approaches $\left.\frac{x^4}{4}\right|_2^4 = 60$ as n increases (the error is 7.5×10^{-3} for $n = 40$). The following code verifies the solution

codes/12.py

2.2.8 Which of the following statements is correct? (CBSE Class 12, 65/7/1, Q.13)

(A) $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$

(B) $\int_{-a}^{a} f(x)\,dx = 0$ if f is even

(C) $\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx$ if f is odd

(D) $\int_0^{2a} f(x)\,dx = \int_0^{a} f(x)\,dx - \int_0^{a} f(2a+x)\,dx$

Solution: **(A)** Put $u = a+b-x$, $du = -dx$. When $x = a$, $u = b$ and when $x = b$, $u = a$, so

$$\int_a^b f(a+b-x)\,dx = \int_b^a f(u)\,(-du) = \int_a^b f(u)\,du \tag{2.2.8.1}$$

(A) is true.
(B) For an even function $\int_{-a}^{a} f = 2\int_0^a f$. With $f(x) = x^2$, $\int_{-1}^{1} x^2\,dx = \frac{2}{3} \neq 0$. False.
(C) For an odd function $\int_{-a}^{a} f = 0$. With $f(x) = x$, the left side is 0 and the right side is 1. False.
(D) The correct splitting is $\int_0^{2a} f = \int_0^a f + \int_0^a f(a+x)\,dx$. With $f(x) = x$, $a = 1$, the left side is 2 and the right side is -2. False.
Hence the correct statement is (A). The program checks each option with SymPy. The following code verifies the solution

codes/13.py

2.2.9 Evaluate $\int \dfrac{e^x}{\sqrt{4-e^{2x}}}\,dx$. (CBSE Class 12, 65/7/1, Q.14)

Solution: Let $u = \frac{e^x}{2}$, so $du = \frac{e^x}{2}\,dx$ and $4 - e^{2x} = 4\left(1-u^2\right)$. Then

$$\int \frac{e^x\,dx}{\sqrt{4-e^{2x}}} = \int \frac{2\,du}{2\sqrt{1-u^2}} = \sin^{-1}u + C = \sin^{-1}\left(\frac{e^x}{2}\right) + C \quad \text{(Option D)} \tag{2.2.9.1}$$

valid for $e^{2x} < 4$, i.e. $x < \ln 2$. To check this, apply the trapezoidal rule over $[-1, 0]$ and compare with $F(0) - F(-1) = \frac{\pi}{6} - \sin^{-1}\left(\frac{e^{-1}}{2}\right) \approx 0.33861$; the rule gives $T_4 = 0.34158$ and $T_{40} = 0.33864$. The following code verifies the solution

codes/14.py

2.2.10 If $\mathbf{a}$ and $\mathbf{b}$ are two vectors such that $|\mathbf{a}+\mathbf{b}| = |\mathbf{a}-\mathbf{b}|$, then $\mathbf{a}$ and $\mathbf{b}$ are (CBSE Class 12, 65/7/1, Q.16)

Solution: (theory only) Squaring both sides and using $|\mathbf{x}|^2 = \mathbf{x}^\top\mathbf{x}$,

$$|\mathbf{a}|^2 + 2\mathbf{a}^\top\mathbf{b} + |\mathbf{b}|^2 = |\mathbf{a}|^2 - 2\mathbf{a}^\top\mathbf{b} + |\mathbf{b}|^2 \tag{2.2.10.1}$$

$$\implies 4\mathbf{a}^\top\mathbf{b} = 0 \tag{2.2.10.2}$$

Hence $\mathbf{a}$ and $\mathbf{b}$ are orthogonal (Option A); the zero vector is the trivial case.

2.2.11 If $\Pr(A) = \frac{1}{7}$, $\Pr(B) = \frac{5}{7}$ and $\Pr(A \cap B) = \frac{4}{7}$, find $\Pr(A' \mid B)$. (CBSE Class 12, 65/7/1, Q.17)

Solution: Let $n = 0$ denote the outcome $A \cap B$ and $n = 1$ the outcome $A' \cap B$, the

two ways in which B can occur. Then

$$p[n] = \Pr(A \cap B)\ \delta[n] + (\Pr(B) - \Pr(A \cap B))\ \delta[n-1] \tag{2.2.11.1}$$

$$= \frac{4}{7}\delta[n] + \frac{1}{7}\delta[n-1] \tag{2.2.11.2}$$

For an event with indicator $m[n]$ and $h[n] = m[-n]$, $\Pr(E) = (p * h)[0]$, the coefficient of z^0 in $P(z)H(z)$, with $P(z) = \frac{4}{7} + \frac{1}{7}z^{-1}$. For B, $m[n] = \delta[n] + \delta[n-1]$ and $H(z) = 1 + z$; for $A' \cap B$, $m[n] = \delta[n-1]$ and $H(z) = z$:

$$\Pr(B) = \left[\left(\frac{4}{7} + \frac{1}{7}z^{-1}\right)(1+z)\right]_{z^0} = \frac{4}{7} + \frac{1}{7} = \frac{5}{7} \tag{2.2.11.3}$$

$$\Pr(A' \cap B) = \left[\left(\frac{4}{7} + \frac{1}{7}z^{-1}\right)z\right]_{z^0} = \frac{1}{7} \tag{2.2.11.4}$$

Hence

$$\Pr(A' \mid B) = \frac{\Pr(A' \cap B)}{\Pr(B)} = \frac{1/7}{5/7} = \frac{1}{5} \tag{2.2.11.5}$$

The values as printed are not consistent with each other, because an intersection cannot be more probable than A alone ($\frac{4}{7} > \frac{1}{7}$); the program prints a warning for this and then evaluates the quantities above. The following code verifies the solution

codes/17.c

2.2.12 A coin is tossed and a card is selected at random from a well-shuffled pack of 52 cards. The probability of getting a head and a face card is (CBSE Class 12, 65/7/1, Q.18)

Solution: Define the indicator maps $X = 1$ for a head and $Y = 1$ for a face card. There are 12 face cards, so the pmfs and their z-transforms are

$$p_X[n] = \frac{1}{2}\delta[n] + \frac{1}{2}\delta[n-1] \implies P_X(z) = \frac{1}{2} + \frac{1}{2}z^{-1} \tag{2.2.12.1}$$

$$p_Y[n] = \frac{10}{13}\delta[n] + \frac{3}{13}\delta[n-1] \implies P_Y(z) = \frac{10}{13} + \frac{3}{13}z^{-1} \tag{2.2.12.2}$$

The two trials are independent, so $S = X + Y$ has pmf $p_S = p_X * p_Y$, i.e. $P_S(z) = P_X(z)P_Y(z)$:

$$P_S(z) = \frac{5}{13} + \frac{13}{26}z^{-1} + \frac{3}{26}z^{-2} \tag{2.2.12.3}$$

and $S = 2$ exactly when both are 1 (a head and a face card). Therefore

$$\Pr(\text{head and face card}) = p_S[2] = \frac{3}{26} \quad \text{(Option B)} \tag{2.2.12.4}$$

The following code verifies the solution

codes/18.c

2.2.13 If $\mathbf{A} = \begin{pmatrix} 2 & 3 \\ -1 & 2 \end{pmatrix}$, show that $\mathbf{A}^2 - 4\mathbf{A} + 7\mathbf{I} = \mathbf{0}$. (CBSE Class 12, 65/7/1, Q.22)

Solution: The trace and determinant are

$$\text{tr}\,(\mathbf{A}) = 2 + 2 = 4, \qquad |\mathbf{A}| = 2\,(2) - 3\,(-1) = 7 \tag{2.2.13.1}$$

The characteristic polynomial of a 2×2 matrix is $\lambda^2 - \text{tr}\,(\mathbf{A})\,\lambda + |\mathbf{A}|$, so

$$\lambda^2 - 4\lambda + 7 = 0 \implies \lambda = \frac{4 \pm \sqrt{16 - 28}}{2} = 2 \pm i\sqrt{3} \tag{2.2.13.2}$$

The eigenvalues sum to 4 (the trace) and multiply to 7 (the determinant). By the Cayley–Hamilton theorem, $\mathbf{A}$ satisfies its own characteristic equation:

$$\mathbf{A}^2 - 4\mathbf{A} + 7\mathbf{I} = \mathbf{0} \tag{2.2.13.3}$$

Direct verification: $\mathbf{A}^2 = \begin{pmatrix} 1 & 12 \\ -4 & 1 \end{pmatrix}$, $4\mathbf{A} = \begin{pmatrix} 8 & 12 \\ -4 & 8 \end{pmatrix}$, $7\mathbf{I} = \begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix}$, so

$$\mathbf{A}^2 - 4\mathbf{A} + 7\mathbf{I} = \begin{pmatrix} 1 - 8 + 7 & 12 - 12 \\ -4 + 4 & 1 - 8 + 7 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \tag{2.2.13.4}$$

The following code verifies the solution

codes/22.c

2.2.14 Differentiate $y = \dfrac{5^x}{x^5}$ with respect to x. (CBSE Class 12, 65/7/1, Q.23(A))

Solution: Write $y = 5^x x^{-5}$ and use the product rule with $\frac{d}{dx}a^x = a^x \ln a$:

$$\frac{dy}{dx} = 5^x \ln 5 \cdot x^{-5} + 5^x \left(-5x^{-6}\right) = \frac{5^x}{x^5}\left(\ln 5 - \frac{5}{x}\right) \tag{2.2.14.1}$$

The following code verifies the solution

codes/23.py

2.2.15 Maximise $Z = 5x + 4y$ subject to $3x + y \le 6$, $x \le 1$, $x \ge 0$, $y \ge 0$. (CBSE Class 12, 65/7/1, Q.24)

Solution: The boundary lines are $3x + y = 6$, $x = 1$, $x = 0$, $y = 0$. The feasible region is bounded by the corner points

$$(0, 0)\,,\ (1, 0)\,,\ (1, 3)\,,\ (0, 6) \tag{2.2.15.1}$$

Corner point (x, y)	$Z = 5x + 4y$
$(0, 0)$	0
$(1, 0)$	5
$(1, 3)$	17
$(0, 6)$	24

TABLE 2.2.15: Value of Z at the corner points

The maximum value is

$$Z_{\max} = 24 \text{ at } (x, y) = (0, 6) \tag{2.2.15.2}$$

The following code verifies the solution

codes/24.c

The following code generates Fig. 2.2.15.1.

codes/24plot.py

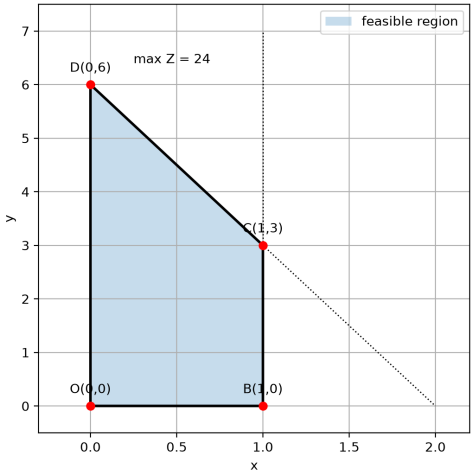


Fig. 2.2.15.1: Feasible region and corner points

2.2.16 A box contains 10 blocks: two marked 0, three marked 1, four marked 2 and one marked 3. A block is drawn at random. Find the probability distribution of the number X on the block and its mean. (CBSE Class 12, 65/7/1, Q.25(A))

Solution: Each block is equally likely, so the pmf is $\Pr(X = x) = \frac{\text{number of blocks marked } x}{10}$:

x	0	1	2	3
$\Pr(X = x)$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{4}{10}$	$\frac{1}{10}$

TABLE 2.2.16: Probability mass function of X

The probabilities sum to 1. In terms of the unit impulse $\delta[n]$ and the z-transform,

$$p[n] = \frac{2\delta[n] + 3\delta[n-1] + 4\delta[n-2] + \delta[n-3]}{10} \tag{2.2.16.1}$$

$$\implies P(z) = \sum_n p[n] z^{-n} = \frac{2 + 3z^{-1} + 4z^{-2} + z^{-3}}{10} \tag{2.2.16.2}$$

Then $P(1) = 1$, and differentiating $P(z)$ gives the mean,

$$E[X] = \sum_n n\,p[n] = -\left.\frac{dP(z)}{dz}\right|_{z=1} = \frac{3+8+3}{10} = \frac{14}{10} = 1.4 \tag{2.2.16.3}$$

The following code verifies the solution

codes/25.c

2.2.17 Two ants move along the lines $2x + 5y - 1 = 0$ and $3x + 2y - 7 = 0$. Using the matrix method, find the point where their paths meet. (CBSE Class 12, 65/7/1, Q.27(A))

Solution: In matrix form,

$$\begin{pmatrix} 2 & 5 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \end{pmatrix} \tag{2.2.17.1}$$

The determinant is $2(2) - 5(3) = -11 \neq 0$, so the solution is unique. By Cramer's rule,

$$x = \frac{1(2) - 5(7)}{-11} = 3, \qquad y = \frac{2(7) - 1(3)}{-11} = -1 \tag{2.2.17.2}$$

The paths meet at $(3, -1)$. The following code verifies the solution

codes/27.c

2.2.18 Find $\frac{dy}{dx}$ if $y = \sqrt{\log\left(\sin\left(\frac{x^3}{3} - 1\right)\right)}$. (CBSE Class 12, 65/7/1, Q.28)

Solution: Let $u = \frac{x^3}{3} - 1$, so $\frac{du}{dx} = x^2$. By the chain rule,

$$\frac{dy}{dx} = \frac{1}{2\sqrt{\log(\sin u)}} \cdot \frac{d}{dx}\log(\sin u) = \frac{1}{2\sqrt{\log(\sin u)}} \cdot \frac{\cos u}{\sin u} \cdot x^2 \tag{2.2.18.1}$$

$$= \frac{x^2 \cot\left(\frac{x^3}{3} - 1\right)}{2\sqrt{\log\left(\sin\left(\frac{x^3}{3} - 1\right)\right)}} \tag{2.2.18.2}$$

The following code verifies the solution

codes/28.py

2.2.19 Minimise $Z = 18x + 10y$ subject to $4x + y \geq 20$, $2x + 3y \geq 30$, $x \geq 0$, $y \geq 0$. (CBSE Class 12, 65/7/1, Q.30)

Solution: The lines $4x + y = 20$ and $2x + 3y = 30$ meet where $y = 20 - 4x$ and $2x + 3(20 - 4x) = 30$, so $x = 3$, $y = 8$. The feasible region is unbounded, with corner points $(0, 20)$, $(3, 8)$ and $(15, 0)$:

$$Z(0, 20) = 200, \qquad Z(3, 8) = 54 + 80 = 134, \qquad Z(15, 0) = 270 \tag{2.2.19.1}$$

The minimum value is

$$Z_{\min} = 134 \text{ at } (3, 8) \tag{2.2.19.2}$$

The following code verifies the solution

codes/30.c

The following code generates Fig. 2.2.19.1.

codes/30plot.py

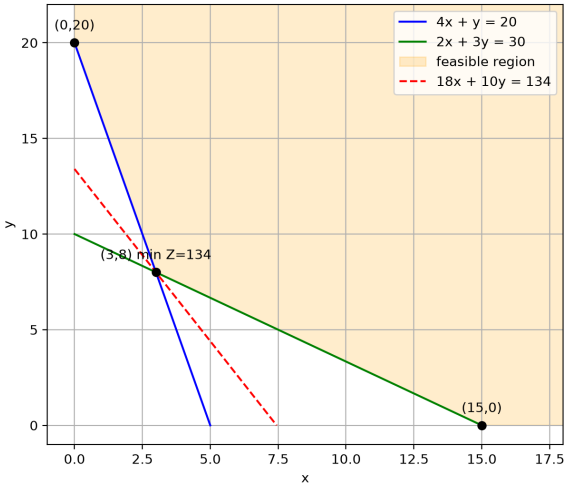


Fig. 2.2.19.1: Unbounded feasible region and the line $Z = 134$

2.2.20 Let $\mathbf{a} = \hat{i} - \hat{j} + 2\hat{k}$, $\mathbf{b} = 2\hat{i} - 4\hat{j} + 5\hat{k}$ and $\mathbf{c} = \lambda\hat{i} - 2\hat{j} - 3\hat{k}$. Find λ if the scalar product of $\mathbf{a}$ with the unit vector along $\mathbf{b} + \mathbf{c}$ is 1. (CBSE Class 12, 65/7/1, Q.31(A))

Solution:

$$\mathbf{b} + \mathbf{c} = \begin{pmatrix} \lambda + 2 \\ -6 \\ 2 \end{pmatrix}, \qquad |\mathbf{b} + \mathbf{c}| = \sqrt{(\lambda + 2)^2 + 40} \tag{2.2.20.1}$$

The scalar product with $\mathbf{a} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ is

$$\frac{\mathbf{a}^\top (\mathbf{b}+\mathbf{c})}{|\mathbf{b}+\mathbf{c}|} = \frac{\lambda+2+6+4}{\sqrt{(\lambda+2)^2+40}} = \frac{\lambda+12}{\sqrt{(\lambda+2)^2+40}} = 1 \tag{2.2.20.2}$$

Squaring,

$$(\lambda+12)^2 = (\lambda+2)^2+40 \implies 20\lambda = -100 \implies \lambda = -5 \tag{2.2.20.3}$$

Check: $\lambda + 12 = 7 > 0$ and $\sqrt{9+40} = 7$, so the unsquared equation holds. The following code verifies the solution

codes/31A.c

2.2.21 Find the shortest distance between the lines $\mathbf{r} = \left(2\hat{i} - \hat{j} + 3\hat{k}\right) + \lambda\left(\hat{i} - 2\hat{j} + 3\hat{k}\right)$ and $\mathbf{r} = \left(\hat{i} + 4\hat{k}\right) + \mu\left(3\hat{i} - 6\hat{j} + 9\hat{k}\right)$. (CBSE Class 12, 65/7/1, Q.31(B))

Solution: The direction vectors $\mathbf{d}_1 = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ and $\mathbf{d}_2 = \begin{pmatrix} 3 \\ -6 \\ 9 \end{pmatrix} = 3\mathbf{d}_1$ are proportional, so the lines are parallel. Take $\mathbf{P}_1 = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and $\mathbf{P}_2 = \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}$, so $\mathbf{P}_2 - \mathbf{P}_1 = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$. The distance between parallel lines is

$$d = \frac{\|(\mathbf{P}_2 - \mathbf{P}_1) \times \mathbf{d}_1\|}{\|\mathbf{d}_1\|} \tag{2.2.21.1}$$

$$\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \\ 1 \end{pmatrix}, \quad d = \frac{\sqrt{25+16+1}}{\sqrt{1+4+9}} = \frac{\sqrt{42}}{\sqrt{14}} = \sqrt{3} \tag{2.2.21.2}$$

The following code verifies the solution

codes/31B.c

2.2.22 Evaluate $\int \frac{x^2+1}{(x^2+2)(2x^2+1)}\,dx$. (CBSE Class 12, 65/7/1, Q.32(A))

Solution: Partial fractions in x^2:

$$\frac{x^2+1}{(x^2+2)(2x^2+1)} = \frac{A}{x^2+2} + \frac{B}{2x^2+1} \implies x^2+1 = A\left(2x^2+1\right) + B\left(x^2+2\right) \tag{2.2.22.1}$$

Comparing coefficients, $2A + B = 1$ and $A + 2B = 1$, so $A = B = \frac{1}{3}$. Using

$\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a}$,

$$\int \frac{x^2+1}{(x^2+2)(2x^2+1)}\,dx = \frac{1}{3\sqrt{2}}\tan^{-1}\left(\frac{x}{\sqrt{2}}\right) + \frac{1}{3\sqrt{2}}\tan^{-1}\left(\sqrt{2}\,x\right) + C \tag{2.2.22.2}$$

Since $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right) + \tan^{-1}\sqrt{2} = \frac{\pi}{2}$, the integral over $[0,1]$ is $\frac{\pi}{6\sqrt{2}} \approx 0.37024$. Applying the trapezoidal rule with step $h = \frac{1-0}{n}$,

$$T_n = \frac{h}{2}\left(f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\right) \tag{2.2.22.3}$$

gives $T_2 = 0.36574$ and $T_{10} = 0.37006$, approaching 0.37024. The following code verifies the solution

codes/32A.py

The following code generates Fig. 2.2.22.1.

codes/32Aplot.py

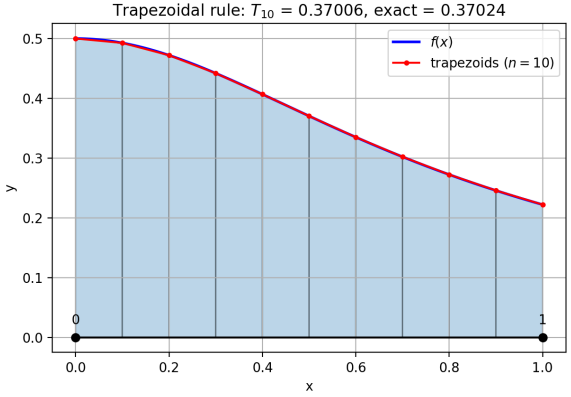


Fig. 2.2.22.1: Trapezoidal rule ($n = 10$) for $f(x)$ on $[0,1]$

2.2.23 Evaluate $\int_0^{\pi} \frac{x\tan x}{\sec x + \tan x}\,dx$. (CBSE Class 12, 65/7/1, Q.32(B))

Solution: Since $\frac{\tan x}{\sec x+\tan x} = \frac{\sin x}{1+\sin x}$, the integral is $I = \int_0^{\pi} \frac{x\sin x}{1+\sin x}\,dx$. Using $\int_0^{\pi} x g(x)\,dx = \frac{\pi}{2}\int_0^{\pi} g(x)\,dx$ when $g(\pi - x) = g(x)$,

$$I = \frac{\pi}{2}\int_0^{\pi} \frac{\sin x}{1+\sin x}\,dx = \frac{\pi}{2}\int_0^{\pi}\left(1 - \frac{1}{1+\sin x}\right)dx \tag{2.2.23.1}$$

Rationalising, $\frac{1}{1+\sin x} = \sec^2 x - \sec x\tan x$, whose antiderivative is $\tan x - \sec x$. Over $[0,\pi]$ this gives $\int_0^{\pi} \frac{dx}{1+\sin x} = 2$ (the point $x = \frac{\pi}{2}$ is handled by taking limits

on either side). Hence

$$I = \frac{\pi}{2}(\pi - 2) \approx 1.7932 \tag{2.2.23.2}$$

Applying the trapezoidal rule to $g(x) = \frac{x \sin x}{1+\sin x}$ on $[0, \pi]$ with $h = \frac{\pi}{n}$ gives $T_{12} = 1.7754$ and $T_{100} = 1.79295$, approaching 1.79321. The following code verifies the solution

codes/32B.py

The following code generates Fig. 2.2.23.1.

codes/32Bplot.py

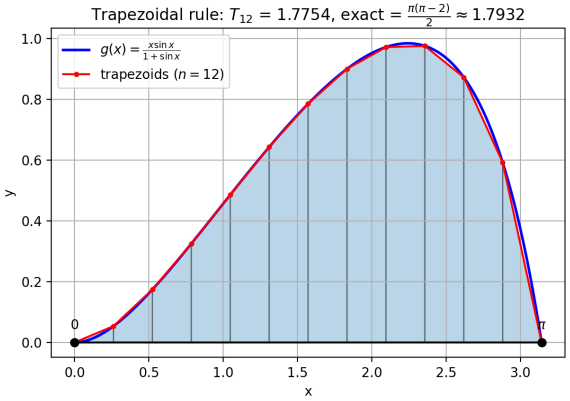


Fig. 2.2.23.1: Trapezoidal rule ($n = 12$) for $\frac{x \sin x}{1+\sin x}$ on $[0, \pi]$

2.2.24 Solve $\frac{dy}{dx} = \cos x - 2y$ using Laplace transforms. (CBSE Class 12, 65/7/1, Q.34)

Solution: Rewrite as $y' + 2y = \cos x$ and let $Y(s) = \mathcal{L}\{y\}$, $y(0) = y_0$. Since $\mathcal{L}\{y'\} = sY - y_0$ and $\mathcal{L}\{\cos x\} = \frac{s}{s^2+1}$,

$$(s+2)Y(s) = y_0 + \frac{s}{s^2+1} \implies Y(s) = \frac{y_0}{s+2} + \frac{s}{(s+2)(s^2+1)} \tag{2.2.24.1}$$

Partial fractions: $\frac{s}{(s+2)(s^2+1)} = \frac{-\frac{2}{5}}{s+2} + \frac{\frac{2}{5}s+\frac{1}{5}}{s^2+1}$. Using $\mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} = e^{-2x}$, $\mathcal{L}^{-1}\left\{\frac{s}{s^2+1}\right\} = \cos x$ and $\mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin x$,

$$y(x) = \left(y_0 - \frac{2}{5}\right)e^{-2x} + \frac{2}{5}\cos x + \frac{1}{5}\sin x = Ce^{-2x} + \frac{2\cos x + \sin x}{5} \tag{2.2.24.2}$$

No initial condition is given, so the constant C is arbitrary. The following code verifies the solution

codes/34.py

2.2.25 Find the points on the line $\frac{2x+4}{6} = \frac{y+1}{2} = \frac{-2z+6}{-4}$ at a distance of $3\sqrt{2}$ units from the point $P\,(1,2,3)$. (CBSE Class 12, 65/7/1, Q.35(A))

Solution: Let the common ratio be t. Then $x = 3t-2$, $y = 2t-1$, $z = 2t+3$. The squared distance from P is

$$(3t-3)^2 + (2t-3)^2 + (2t)^2 = 17t^2 - 30t + 18 = 18 \tag{2.2.25.1}$$

$$t\,(17t-30) = 0 \implies t = 0 \text{ or } t = \frac{30}{17} \tag{2.2.25.2}$$

The required points are

$$Q_1 = (-2,-1,3)\,, \qquad Q_2 = \left(\frac{56}{17}, \frac{43}{17}, \frac{111}{17}\right) \tag{2.2.25.3}$$

The following code verifies the solution

codes/35A.c

The following code generates Fig. 2.2.25.1.

codes/35Aplot.py

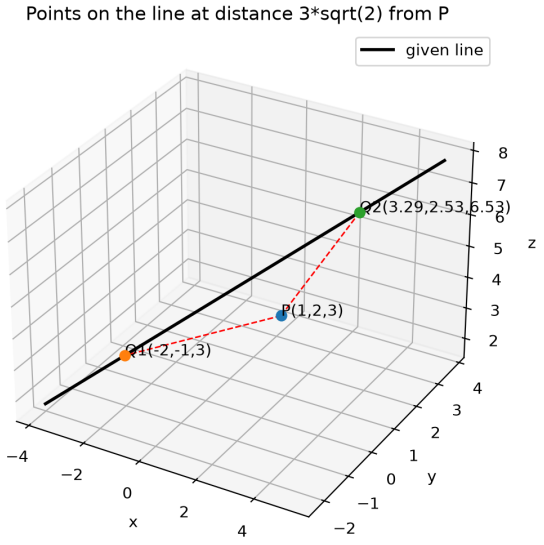


Fig. 2.2.25.1: Points Q_1, Q_2 on the line at distance $3\sqrt{2}$ from P

2.2.26 Find the image of the point $P\left(-1,5,2\right)$ in the line $\frac{2x-4}{2}=\frac{y}{2}=\frac{2-z}{3}$, and the length PP'. (CBSE Class 12, 65/7/1, Q.35(B))

Solution: Let the common ratio be t: $x=t+2$, $y=2t$, $z=2-3t$. The line passes through $\mathbf{A}=\begin{pmatrix}2\\0\\2\end{pmatrix}$ with direction $\mathbf{d}=\begin{pmatrix}1\\2\\-3\end{pmatrix}$. The foot of the perpendicular $H=\mathbf{A}+t_0\mathbf{d}$ has

$$t_0=\frac{\left(\mathbf{P}-\mathbf{A}\right)^\top\mathbf{d}}{\mathbf{d}^\top\mathbf{d}}=\frac{\begin{pmatrix}-3\\5\\0\end{pmatrix}^\top\begin{pmatrix}1\\2\\-3\end{pmatrix}}{14}=\frac{7}{14}=\frac{1}{2}\implies\mathbf{H}=\begin{pmatrix}\frac{5}{2}\\1\\\frac{1}{2}\end{pmatrix}\tag{2.2.26.1}$$

Since H is the mid-point of PP',

$$\mathbf{P}'=2\mathbf{H}-\mathbf{P}=\begin{pmatrix}5\\2\\1\end{pmatrix}-\begin{pmatrix}-1\\5\\2\end{pmatrix}=\begin{pmatrix}6\\-3\\-1\end{pmatrix}\tag{2.2.26.2}$$

$$PP'=\sqrt{7^2+8^2+3^2}=\sqrt{122}\tag{2.2.26.3}$$

The following code verifies the solution

codes/35B.c

The following code generates Fig. 2.2.26.1.

codes/35Bplot.py

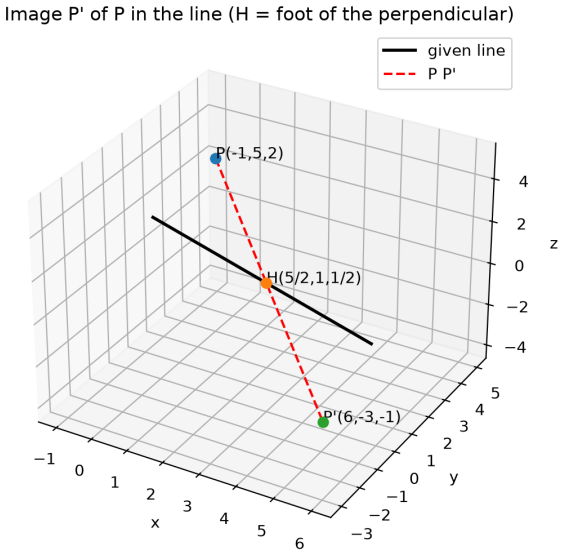


Fig. 2.2.26.1: Point P, its image P' and the foot H

2.2.27 **Case study (camphor).** The radius r of a camphor tablet decreases with time t (in hours) as it evaporates, and $\frac{dr}{dt} = -\frac{2}{3}k$ with $k > 0$ and $r(0) = 5$ mm. (i) Write the order and degree of the differential equation. (ii) Solve it using the Laplace transform. (iii)(a) If $r = 3$ mm after 1 hour, find the time for the radius to become 0. **OR** (iii)(b) If $r = 1$ mm after 1 hour, find the time for the radius to become 0. (CBSE Class 12, 65/7/1, Q.37)

Solution: **(i)** The highest derivative is $\frac{dr}{dt}$ and its power is 1, so the order is 1 and the degree is 1.

(ii) Let $R(s) = \mathcal{L}\{r\}$. Then

$$sR(s) - r(0) = -\frac{2k}{3s} \implies R(s) = \frac{5}{s} - \frac{2k}{3s^2} \implies r(t) = 5 - \frac{2}{3}kt \quad (2.2.27.1)$$

(iii)(a) From $r(1) = 3$: $3 = 5 - \frac{2}{3}k \implies k = 3$, so $r(t) = 5 - 2t$. Setting $r = 0$,

$$t = \frac{5}{2} \text{ hours} \quad (2.2.27.2)$$

(iii)(b) From $r(1) = 1$: $1 = 5 - \frac{2}{3}k \implies k = 6$, so $r(t) = 5 - 4t$ and

$$t = \frac{5}{4} \text{ hours} \quad (2.2.27.3)$$

The following code verifies the solution

codes/37.c

2.2.28 **Case study (health survey).** Of 1000 people, 700 have good health (A_1), 200 average

health (A_2) and 100 poor health (A_3). The probabilities of contracting the disease D are $\Pr(D \mid A_1) = 0.25$, $\Pr(D \mid A_2) = 0.35$ and $\Pr(D \mid A_3) = 0.50$. (i) Find the probability that a randomly selected person has contracted the disease. (ii) Find the probability that a person who has not contracted the disease belongs to category A_1. (CBSE Class 12, 65/7/1, Q.38)

Solution: Let $n = 0, 1, 2$ denote the categories A_1, A_2, A_3, and let $d[n] = \Pr(D \mid A_{n+1})$. Then

$$p[n] = 0.7\,\delta[n] + 0.2\,\delta[n-1] + 0.1\,\delta[n-2] \tag{2.2.28.1}$$

$$d[n] = 0.25\,\delta[n] + 0.35\,\delta[n-1] + 0.50\,\delta[n-2] \tag{2.2.28.2}$$

(i) By the law of total probability, with $h[n] = d[-n]$,

$$\Pr(D) = \sum_n p[n]\,d[n] = (p * h)\,[0] \tag{2.2.28.3}$$

which is the coefficient of z^0 in $P(z)\,H(z)$, where $P(z) = 0.7 + 0.2z^{-1} + 0.1z^{-2}$ and $H(z) = 0.25 + 0.35z + 0.50z^2$:

$$\Pr(D) = 0.7\,(0.25) + 0.2\,(0.35) + 0.1\,(0.50) = 0.295 \tag{2.2.28.4}$$

(ii) For D' the conditional pmf is $d'[n] = 1 - d[n] = 0.75,\ 0.65,\ 0.50$, so in the same way $\Pr(D') = 1 - 0.295 = 0.705$. For the numerator, restrict $p[n]$ to A_1, i.e. $p_1[n] = 0.7\,\delta[n]$, and let $h'[n] = d'[-n]$:

$$\Pr(A_1 \cap D') = (p_1 * h')\,[0] = 0.7\,(0.75) = 0.525 \tag{2.2.28.5}$$

By Bayes' theorem,

$$\Pr(A_1 \mid D') = \frac{\Pr(A_1 \cap D')}{\Pr(D')} = \frac{0.525}{0.705} = \frac{35}{47} \approx 0.7447 \tag{2.2.28.6}$$

The following code verifies the solution

codes/38.c